

pre-defined and signed by Google so any non-authorised executable bit is purged if not verified by the device. This verified booting takes place to repair and resume operations in a seamless fashion for the user. Even in the case of bit rot (diminishing software performance) the verified boot switches to repair the operating system to its most recent version.

The customized firmware is further encrypted by Google's 8192-bit public key which is hard wired into the device at the factory stage which ensures that the PKI (public key infrastructure) chain of trust verifies the operating system as being untampered. This verification takes place on every start-up since the first part of the BIOS flash that is accessed is read-only. This read-only BIOS verifies the read-write BIOS and then executes it. The next step in the chain is the read-write BIOS verifying the kernel and executing that, which then verifies the roots of every block as it comes off the drive. This secured chain resets to Recovery mode if any of these steps is tampered with. In ideal conditions the whole start-up boot process to login takes seven seconds.

This extremely fast and secure booting allows the system to practically stay always on - which is how most people treat their online surfing habits. The added stability brought on by remote operating system upgrades allows the Chrome OS to be the perfect medium for data sensitive activities like online banking. It is also perfectly suited to the enterprise work environment where customised apps or subscription corporate apps are favoured to retain security within an organisation. In effect, the Chrome OS solves many of the Bring Your Device To Work problems by relying on Google's security features.

Chromebook, Chrome OS or Chromium OS?

We will go on to explain in later chapters how you can experience both Google's version of Chrome OS as well as the developer friendly open source Chromium OS. But it's abundantly clear that the true power and selling point of the Chrome OS can't be replicated fully when used as an upgrade to Chromium OS. The value of the Chrome OS is directly tied into two key features - internet connectivity and firmware design.

As hobbyists or experimental users, you can get a very accurate sense of what Chrome OS is all about by using it on your own devices. The immediate results themselves can be surprising to most users, since Chrome OS consolidates your user experience in a brand new way in to one window while giving you the familiar flexibility to play local media. As a casual user